

CHAPTER 1: DATABASE MANAGEMENT SYSTEM (DBMS)

SHORT ANSWER QUESTIONS (5 Marks)

1. What is normalization? Explain 2NF and/or 3NF with examples.

Asked: 9 times

(2082 Q.No.10OR | 2081 GIE Set A Q.No.10 | 2081 Q.No.10OR | 2080 GIE Set B Q.No.10 | 2079 Set A Q.No.10OR | 2077 Set D Q.No.4 | 2076 GIE Set A Q.No.7 | 2076 GIE Set B Q.No.7 | 2074 Set A Q.No.8)

2. Explain the Relational Database Model with example.

Asked: 4 times

(2080 GIE Set A Q.No.10OR | 2080 Q.No.10 | 2079 GIE Set A Q.No.10 | 2073 Set D Q.No.9)

3. Differentiate between centralized and distributed database system.

Asked: 4 times

(2081 GIE Set B Q.No.10OR | 2079 Set A Q.No.10 | 2076 GIE Set B Q.No.10 | 2075 Set A Q.No.8)

4. What are the roles/functions of a Database Administrator (DBA)? Explain.

Asked: 3 times

(2079 GIE Set B Q.No.10 | 2078 Set C Q.No.10 | 2076 GIE Set B Q.No.8)

5. Write any three differences between DDL and DML. Give examples of each.

Asked: 2 times

(2082 Q.No.10 | 2076 Q.No.8)

6. Describe DDL statements with examples in SQL.

Asked: 2 times

(2079 GIE Set A Q.No.10OR | 2079 GIE Set B Q.No.10OR)

7. Write SQL DDL statements to create a table with given schema/attributes.

Asked: 2 times

(2081 GIE Set A Q.No.10OR | 2080 Q.No.10OR)

8. List the different types of database models. Explain Relational Database model in brief.

Asked: 2 times

(2081 GIE Set B Q.No.10 | 2078 Set C Q.No.8)

9. Evaluate the advantages/importance of DBMS compared to traditional file-based systems.

Asked: 2 times

(2081 Q.No.10 | 2080 GIE Set A Q.No.10)

10. Explain E-R diagram / entity, attribute and relationship in DBMS.

Asked: 2 times

(2078 Set C Q.No.9 | 2076 GIE Set A Q.No.8)

11. What is the importance of database security in DBMS? Describe.

Asked: 1 time

(2080 GIE Set B Q.No.10OR)

12. Explain the different types of data integrity in DBMS.

Asked: 1 time

(2076 Q.No.7)

LONG ANSWER QUESTIONS (8 Marks)

1. Define normalization. Explain 2NF and 3NF with examples. / What is normalization? Write its advantages and explain 2NF and 3NF with examples.

Asked: 3 times

(2078 Set C Q.No.4 | 2075 Q.No.5 | 2075 Set B Q.No.5)

2. What do you understand by data integrity? Why is it important while designing a database? State and describe different types of data integrity.

Asked: 1 time

(2062 Q.No.2OR)

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CHAPTER 2: DATA COMMUNICATION AND NETWORKING

SHORT ANSWER QUESTIONS (5 Marks)

1. What is transmission media? Describe/explain the types of transmission media with examples.

Asked: 6 times

(2076 GIE Set B Q.No. 9 | 2075 Set A Q.No. 10 | 2075 Set B Q.No. 10 | 2074 Set A Q.No. 10 | 2074 Set B Q.No. 9 | 2071 Set C Q.No. 10)

2. Compare/differentiate star and ring topology with features/pros and cons.

Asked: 2 times

(2079 GIE Set B Q.No. 11 OR | 2077 Set D Q.No. 5)

3. Define network topology. Explain the Ring topology with diagram/features.

Asked: 2 times

(2076 Q.No. 9 | 2075 GIE Q.No. 9)

4. Differentiate between LAN and WAN.

Asked: 2 times

(2076 GIE Set A Q.No. 10 | 2074 Set B Q.No. 10)

5. Differentiate between peer-to-peer and client/server network.

Asked: 2 times

(2075 Set A Q.No. 9 | 2073 Supp Q.No. 9)

6. Describe simplex, half duplex, and full duplex with examples.

Asked: 2 times

(2072 Set C Q.No. 11 | 2070 Supp Q.No. 10)

7. What is Networking? List the advantages of Networking.

Asked: 2 times

(2072 Set D Q.No. 9 | 2070 Set C Q.No. 8)

8. Differentiate between bounded and unbounded transmission media.

Asked: 1 time

(2075 GIE Q.No. 10)

9. Describe the Bus and Star topology with suitable diagram.

Asked: 1 time

(2073 Set C Q.No. 9)

10. What is computer network? List out hardware equipments to establish the computer network.

Asked: 1 time

(2076 GIE Set A Q.No. 9)

11. Describe the wireless network system. List out devices and equipments necessary for Wi-Fi network.

Asked: 1 time

(2076 Q.No. 10)

12. Describe the class C IP address with an example.

Asked: 1 time

(2079 GIE Set B Q.No. 11)

13. Explain about coaxial cable and fiber optics cable.

Asked: 1 time

(2078 Set C Q.No. 11)

LONG ANSWER QUESTIONS (8 Marks)

1. Explain client-server and peer-to-peer network architecture with diagram. Write advantages and disadvantages of each.

Asked: 2 times

(2081 GIE Set B Q.No. 15 | 2079 GIE Set A Q.No. 15)

2. Compare star and bus network topology. Which network cable is suitable for the school's/local area network? Justify.

Asked: 2 times

(2081 Q.No. 15 | 2080 GIE Set B Q.No. 15)

3. What is transmission medium? Explain its major types with advantages and disadvantages.

Asked: 1 time

(2082 Q.No. 15)

4. Why do network designers prefer switch in LAN? Why do people prefer star topology in organizations like a bank? Give reasons.

Asked: 1 time

(2081 GIE Set A Q.No. 15)

5. Describe CAT-6 and optical fiber cable features. Which is more suitable to develop the local area network?

Asked: 1 time

(2080 GIE Set A Q.No. 15)

6. What is network topology? Differentiate between guided (wire) and unguided (wireless) media with examples.

Asked: 1 time

(2080 Q.No. 15)

7. Compare star and ring topology with pros and cons. Which data communication cable is more appropriate to design a local area network? Describe.

Asked: 1 time

(2079 Set A Q.No. 15)

SCROLL DOWN

CHAPTER 3: WEB TECHNOLOGY II

SHORT ANSWER QUESTIONS (5 Marks)

1. Write a PHP script/program to connect to a MySQL database / explain database connection method in PHP / explain the mysqli_connect() function.

Asked: 7 times

(2082 Q.No.11OR | 2081 Q.No.11OR | 2080 GIE Set A Q.No.11OR | 2080 GIE Set B Q.No.11OR | 2080 Q.No.11 | 2079 GIE Set A Q.No.11OR | 2079 Set A Q.No.11OR)

2. Write a JavaScript program to find the largest/smallest number among three numbers.

Asked: 3 times

(2082 Q.No.11 | 2081 GIE Set B Q.No.11 | 2080 GIE Set A Q.No.11)

3. Write a JavaScript program to find the factorial of any given number.

Asked: 2 times

(2080 Q.No.11OR | 2079 Set A Q.No.11)

4. Write a JavaScript code/function that displays the multiplication table of an entered number.

Asked: 1 time

(2081 GIE Set A Q.No.11)

5. Develop a PHP code to add/insert a record into a database table.

Asked: 1 time

(2081 GIE Set A Q.No.11OR)

6. Write a PHP program to swap two numbers.

Asked: 1 time

(2081 GIE Set B Q.No.11OR)

7. Write a JavaScript function that checks if a number is even or odd and print the result.

Asked: 1 time

(2081 Q.No.11)

8. How do you add an event handler in JavaScript? Give an example.

Asked: 1 time

(2080 GIE Set B Q.No.11)

9. Write a JavaScript function to find the sum of two numbers.

Asked: 1 time

(2079 GIE Set A Q.No.11)

10. How do you fetch data from a database in PHP and display it in a form?

Asked: 1 time

(2079 Set A Q.No.11OR)

11. Differentiate between server-side and client-side scripting.

Asked: 1 time

(No year tag)

12. What is PHP? Explain its uses and advantages. Write basic PHP syntax with an example program.

Asked: 1 time

(No year tag)

13. Explain different data types used in JavaScript.

Asked: 1 time

(No year tag)

14. What is JQuery? Write down its uses.

Asked: 1 time

(No year tag)

15. Explain the different operators used in PHP.

Asked: 1 time

(No year tag)

LONG ANSWER QUESTIONS (8 Marks)

1. What are the uses of JavaScript in web page development? How do you connect PHP and MySQL database? Explain with connection class.

Asked: 1 time

(2079 GIE Set B Q.No.15)

2. Write a server-side script to create a database, connect with it, create a table and insert data into it.

Asked: 1 time

(No year tag)

3. What is object-based programming? Explain different JavaScript objects.

Asked: 1 time

(No year tag)

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CHAPTER 4: PROGRAMMING IN C

LONG ANSWER QUESTIONS (8 Marks)

1. Write a C program using structure to store and display employees/students records (name, salary, post, age, address, etc.).

Asked: 20+ times

(2082 Q.No.16 | 2081 GIE Set B Q.No.16 | 2081 Q.No.16 | 2080 GIE Set A Q.No.16 | 2080 GIE Set B Q.No.16 | 2080 Q.No.16OR | 2079 GIE Set A Q.No.16 | 2079 Set A Q.No.16 | 2078 Set C Q.No.5 | 2076 GIE Set B Q.No.5 | 2076 Q.No.4 | 2075 GIE Q.No.4 | 2075 Set A Q.No.4 | 2075 Set B Q.No.4 | 2074 Supp Q.No.5 | 2074 Set A Q.No.5 | 2074 Set B Q.No.5 | 2073 Set C Q.No.5 | 2066 Q.No.2a | 2061 Q.No.1b)

2. Write a C program to store data of employees/students (name, address, salary, roll, marks etc.) into a file and read/display records from the file.

Asked: 16+ times

(2081 GIE Set B Q.No.16OR | 2080 GIE Set A Q.No.16OR | 2080 GIE Set B Q.No.16OR | 2079 GIE Set A Q.No.16OR | 2079 GIE Set B Q.No.16OR | 2079 Set A Q.No.16OR | 2076 GIE Set A Q.No.5 | 2073 Supp Q.No.5b | 2073 Set D Q.No.5a | 2072 Set C Q.No.5 | 2072 Set D Q.No.5 | 2071 Set D Q.No.5 | 2070 Supp Q.No.5 | 2070 Set D Q.No.5 | 2068 Q.No.3b | 2066 Q.No.3a | 2065 Q.No.3a | 2062 Q.No.3b)

3. What is function? Write a C program to find factorial of a given number using recursive/user-defined function.

Asked: 8 times

(2079 GIE Set B Q.No.16 | 2076 GIE Set B Q.No.3 | 2071 Set C Q.No.1 | 2070 Set D Q.No.4 | 2067 Q.No.1b | 2063 Q.No.1b | 2082 Q.No.16OR | 2081 Q.No.16OR)

4. Write a C program using function to check whether a given number is odd or even.

Asked: 4 times

(2076 GIE Set A Q.No.3 | 2074 Supp Q.No.3 | 2074 Set B Q.No.3 | 2074 Set A Q.No.4 / 2073 Set C Q.No.4)

5. Differentiate between structure and union with syntax / Write a program using structure and union.

Asked: 4 times

(2073 Set C Q.No.5 | 2072 Set D Q.No.4a | 2065 Q.No.2b | 2078 Set C Q.No.5)

6. What is pointer? Describe the benefits/advantages of pointer with examples.

Asked: 4 times

(2078 Set C Q.No.2a | 2077 Set D Q.No.3a | 2071 Set D Q.No.3 | 2068 Set D Q.No.3)

7. What are the components of a function in C? Describe call-by-value and call-by-reference / passing function parameters.

Asked: 3 times

(2082 Q.No.16OR | 2076 GIE Set A Q.No.3 | 2067 Q.No.2b)

8. Write a C program using function to find the sum of n integer numbers.

Asked: 3 times

(2074 Set A Q.No.4 | 2073 Set C Q.No.4 | 2066 Q.No.2b)

9. Describe fscanf() and fprintf() functions. Write a program using these file handling functions.

Asked: 3 times

(2072 Set C Q.No.5 | 2071 Supp Q.No.5 | 2080 GIE Set B Q.No.16OR)

10. Describe file handling modes in C. Write a C program to create and write data into a file.

Asked: 3 times

(2081 Q.No.16OR | 2080 GIE Set B Q.No.16OR | 2079 GIE Set B Q.No.16OR)

11. Write a C program using structure to store student records and sort/rearrange names alphabetically.

Asked: 2 times

(2066 Q.No.2a | 2061 Q.No.1b)

12. Write a C program using function to find the simple interest / multiply two integer numbers.

Asked: 2 times

(2075 Set B Q.No.3 | 2073 Supp Q.No.4a)

13. What is recursion? Write a C program to calculate factorial using recursive function.

Asked: 2 times

(2070 Set D Q.No.4 | 2071 Set C Q.No.1)

14. Write a C program using user-defined function to find the area of a shape (pond, football, etc.).

Asked: 2 times

(2080 Q.No.16 | 2076 Q.No.1)

15. Write a C program to store 10 student records with roll number, name and marks and display them using array and structure.

Asked: 2 times

(2082 Q.No.16 | 2081 GIE Set A Q.No.16OR)

16. What is binary file handling in C? Explain how putw() and getw() / fprintf() and fscanf() facilitate binary input/output operations.

Asked: 2 times

(2081 Q.No.16OR | 2081 GIE Set B Q.No.16OR)

17. Define control structure. Write a C program to find sum, difference and greater of two integers using switch case.

Asked: 1 time

(2073 Supp Q.No.3)

18. Write a C program to enter name, roll-no and marks of 10 students, store in file, read and display from the file.

Asked: 2 times

(2068 Q.No.3b | 2079 Set A Q.No.16OR)

19. Write a C program to delete and rename a data file using remove() and rename() commands.

Asked: 1 time

(2064 Q.No.3a)

20. What is the difference between library function and user-defined function? Write a program to calculate y raised to power x using user-defined function.

Asked: 1 time

(2067 Q.No.2b)

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CHAPTER 5: OBJECT-ORIENTED PROGRAMMING (OOP)

SHORT ANSWER QUESTIONS (5 Marks)

1. Explain the terms polymorphism and inheritance in OOP with examples.

Asked: 4 times

(2071 Supp Q.No.14 | 2067 Q.No.9 | 2063 Q.No.5 | 2061 Q.No.11)

2. Distinguish between OOP and procedural programming language / Compare OOP and procedural programming.

Asked: 3 times

(2080 Q.No.12 | 2079 Set A Q.No.12 | 2075 GIE Q.No.11)

3. What is inheritance in OOP? Describe/explain with an example.

Asked: 3 times

(2082 Q.No.12 | 2080 GIE Set A Q.No.12 | 2079 GIE Set B Q.No.12)

4. Describe class and object in OOP with a real-world example.

Asked: 3 times

(2081 Q.No.12 | 2080 GIE Set B Q.No.12 | 2081 GIE Set A Q.No.12)

5. What are the importance/advantages of OOP?

Asked: 3 times

(2079 GIE Set A Q.No.12 | 2075 Set A Q.No.11 | 2071 Set D Q.No.11)

6. Define "Class". Explain encapsulation and how it helps in data hiding in OOP.

Asked: 2 times

(2081 GIE Set A Q.No.12 | 2081 GIE Set B Q.No.12)

7. What is OOP? Write the application / list the characteristics of OOP.

Asked: 2 times

(2073 Supp Q.No.10 | 2072 Set D Q.No.12)

SCROLL DOWN

CHAPTER 6: SOFTWARE PROCESS MODEL

SHORT ANSWER QUESTIONS (5 Marks)

1. Explain SDLC / describe the stages of SDLC / explain the importance of SDLC in software development / explain system analysis phase of SDLC.

Asked: 10+ times

(2081 GIE Set A Q.No.13 | 2081 GIE Set B Q.No.13 | 2080 Q.No.13 | 2079 GIE Set A Q.No.13 | 2079 Set A Q.No.13 | 2076 GIE Set B Q.No.6 | 2076 Q.No.6 | 2075 GIE Q.No.6 | 2075 Set B Q.No.6 | 2070 Supp Q.No.6 | 2069 Q.No.6 | 2063 Q.No.4 | 2059 Q.No.9)

2. Who is a system analyst? Explain the roles/responsibilities/characteristics of a system analyst in SDLC.

Asked: 10 times

(2078 Set C Q.No.7 | 2076 GIE Set A Q.No.6 | 2075 Set A Q.No.6 | 2075 Set B Q.No.7 | 2073 Set D Q.No.6 | 2068 Cancelled Q.No.8 | 2067 Q.No.4 | 2065 Q.No.5 | 2064 Q.No.6 | 2057 Q.No.2)

3. What is feasibility study? Explain / Explain different levels of feasibility study / Why is it necessary before designing a system?

Asked: 9 times

(2078 Set C Q.No.6 | 2072 Set E Q.No.6 | 2071 Supp Q.No.6 | 2070 Set C Q.No.6 | 2068 Q.No.5 | 2067 Q.No.9 | 2064 Q.No.9 | 2062 Q.No.5 | 2060 Q.No.4)

4. What is requirement gathering? Explain different requirement gathering/collection methods/techniques.

Asked: 4 times

(2082 Q.No.13 | 2081 Q.No.13 | 2080 GIE Set A Q.No.13 | 2079 GIE Set A Q.No.13)

5. Define the terms DFD and E-R diagram / What is E-R diagram? Write the meaning of graphical symbols used in E-R diagram.

Asked: 4 times

(2075 GIE Q.No.7 | 2075 Set A Q.No.7 | 2075 Set B Q.No.9 | 2073 Supp Q.No.7)

6. What are the documentation techniques? Explain / What is documentation? Explain its importance in program designing.

Asked: 4 times

(2068 Cancelled Q.No.10 | 2065 Q.No.11 | 2061 Q.No.9 | 2037 Q.No.9)

7. What are the different program logic tools? Explain decision table and decision tree / flowchart symbols with examples.

Asked: 4 times

(2068 Q.No.7 | 2066 Q.No.9 | 2063 Q.No.11 | 2066 Q.No.11)

8. Explain testing techniques during system development / How can you test a newly developed system?

Asked: 3 times

(2066 Q.No.11 | 2066 Supp Q.No.4 | 2080 GIE Set B Q.No.13)

9. Describe the waterfall software development model with pros and cons / Explain agile software development methodology.

Asked: 2 times

(2079 GIE Set B Q.No.13 | 2080 GIE Set B Q.No.13)

10. What are the major activities performed to design the software? Describe.

Asked: 1 time

(2079 Set A Q.No.13)

11. State the steps involved in system development cycle. Explain data security laws.

Asked: 1 time

(2059 Q.No.9)

SCROLL DOWN

CHAPTER 7: RECENT TRENDS IN TECHNOLOGY

SHORT ANSWER QUESTIONS (5 Marks)

1. What is AI? Explain application areas / roles of AI in real life / how AI affects modern society.

Asked: 8 times

(2082 Q.No.14 | 2081 Q.No.14 | 2080 GIE Set B Q.No.14 | 2079 Set A Q.No.14 | 2077 Set D Q.No.7 | 2076 Q.No.14 | 2068 Q.No.11 | 2061 Q.No.12)

2. What is e-governance? List out objectives / importance of e-governance / key challenges of implementing e-governance.

Asked: 6 times

(2072 Set C Q.No.13 | 2071 Supp Q.No.11 | 2071 Set D Q.No.14 | 2070 Set D Q.No.14 | 2069 Q.No.11 | 2079 GIE Set B Q.No.14)

3. What is E-commerce? Explain advantages / role to enhance digital economy / list demerits of E-commerce.

Asked: 4 times

(2078 Set C Q.No.14 | 2075 Set B Q.No.14 | 2066 Supp Q.No.7 | 2079 GIE Set B Q.No.14)

4. Define the terms e-learning and virtual reality / e-business and e-learning / e-governance and e-medicine. Write the importance of multimedia in e-learning.

Asked: 4 times

(2076 GIE Set B Q.No.13 | 2075 GIE Q.No.14 | 2075 Set A Q.No.12 | 2073 Set D Q.No.14)

5. What is cloud computing? Explain concept / advantages and disadvantages / define cloud computing services.

Asked: 3 times

(2080 GIE Set A Q.No.14 | 2080 Q.No.14 | 2079 GIE Set A Q.No.14)

6. Define Robotics. Explain the application of Robotics in various fields such as industry and healthcare.

Asked: 1 time

(2081 GIE Set A Q.No.14)

7. Define the terms Big Data and IoT (Internet of Things).

Asked: 1 time

(2081 GIE Set B Q.No.14)

8. What are the advantages and disadvantages of social media? List out.

Asked: 1 time

(2074 Set B Q.No.13)

9. How can you minimize computer crime? Explain.

Asked: 1 time

(2078 Set C Q.No.13)

10. What is AI? Explain cyber law.

Asked: 1 time

(2073 Set D Q.No.12)